

# Titanic Data Analysis Report

## 1. Summary Statistics

The dataset includes 331 complete entries with the following stats:

- Passenger class Mean = 2.14 (1st to 3rd).
- Age Mean = 30.18, range = 0.17-76.
- Most passengers were alone.
- Fare: Mean = 40.98, heavily skewed with high outliers up to 512.

## 2. Box Plot Observations

Age is mostly symmetric with some infant outliers.

## 3. Histogram Observation

Age mostly between 20-40.

## 4. Scatter Plot / Pairplot Trends

- Fare vs. Age: Slight positive trend.

## 5. Heatmap Insights (Correlation Matrix)

- Passenger class & Fare: -0.59 -> Lower class = cheaper fare.
- Passenger class & Age: -0.50 -> Lower class = younger passengers.
- Age & Fare: +0.34 -> Older people paid more.

## Overall Summary

- Passenger class greatly affects fare and age.
- Fare is skewed with strong class-based differences.
- Most passengers were alone, with low Siblings/ spouse and Parents /children.
- Moderate correlations found between family size and fare/age.
- Dataset contains meaningful trends for further analysis.